

EDUCATION

Stanford University	Sep 2018 - Jun 2022
◦ B.S., Computer Science (AI Specialization) • GPA: 4.18 / 4.0 • Graduated with Distinction (Summa Cum Laude) ◦ Endorsed by Fields Medalist Prof. Ngo Bao Chau for exceptional mathematical talent and research potential.	

AWARDS

International Mathematical Olympiad: Bronze Medal	American Mathematics Competition: Gold Medal
Vietnam Math Olympiad: Gold Medal (Rank #1)	International Math Local Tournament: Gold Medal
Hanoi Open Math Competition: Gold Medal	Regional Math Competitions: 5+ Gold Medals

WORK EXPERIENCE

Meta Platforms	Senior Machine Learning Engineer	July 2022 - Present
◦ Ranked #1 for organizational impacts (Exceeded Expectations+), delivering 90-120% of revenue targets and 40-50% of DAU goals through development and ownership of high-leverage ML solutions. ◦ Architected end-to-end post-training and serving pipelines for compact LLMs, optimizing fine-tuning and alignment for high-scale generation and ranking. Engineered robust LLM-based guardrails to preemptively neutralize safety, integrity, and quality risks, enforcing a critical pre-deployment defense layer. ◦ Pioneered next-generation modeling architectures by developing novel reinforcement learning and causal inference technologies for multi-objective and long-term user value optimization. Advanced model capabilities through knowledge distillation, Transformer-based long-horizon sequence modeling, and contextual representation learning, capturing deep latent interests to drive engagement. ◦ Led large-scale, cross-organizational initiatives spanning 20+ engineers across infrastructure, product, and modeling teams. Independently defined strategies, aligned stakeholders, and executed system-wide upgrades that materially advanced Meta's notification recommendation system. ◦ Scaled team capability by 3x by proactively scoping and launching new technical roadmaps, mentoring and onboarding 10+ IC4/IC5 engineers, and enabling sustained growth and research-grade innovation.		
Meta Platforms	Software Engineering Intern	Jun 2021 - Sep 2021
◦ Developed large-scale ML models in PyTorch to generate contextual embeddings of users and ads, enhancing semantic understanding and fine-grained personalization in ad retrieval and ranking systems. ◦ Designed and deployed representation learning frameworks for user-ad matching across multiple surfaces, significantly improving relevance and driving ad revenue growth. Used Python, SQL, C++, PHP, and Spark.		
Meta Platforms	Software Engineering Intern	Jun 2020 - Sep 2020
◦ Enhanced latency profiling tools with module-level debugging. Achieved a 5x increase in runtime efficiency. ◦ Accelerated Conv1D and channel shuffle operations via low-level kernel optimizations for PyTorch, delivering up to 10x operator-level speedup for on-device speech and NLP models. Used Python and C++.		

RESEARCH EXPERIENCE

Stanford AI Lab	Undergraduate Researcher	Apr 2020 - Sep 2021
◦ Built large-scale data pipelines in Python, NumPy, and Pandas to ingest, align, and preprocess satellite imagery and global forest loss driver labels. Implemented advanced data augmentation and stratified sampling techniques to improve signal diversity and model robustness across diverse geographies. ◦ Designed and trained deep learning models in PyTorch, including CNNs, LSTMs, and multimodal fusion architectures, to classify forest loss drivers from multi-temporal satellite imagery, achieving 80% classification accuracy and supporting scalable environment monitoring.		
Stanford InfoLab	Undergraduate Researcher	Jan 2020 - Apr 2020
◦ Engineered a high-throughput input pipeline for loading and preprocessing underwater video data, including image normalization, spatial augmentation, and temporal slicing, to support robust model training. ◦ Developed Mask R-CNN and U-Net models in TensorFlow to detect, localize, and temporally track coral structures in underwater environments, enabling fine-grained analysis and monitoring of reef health over time.		
Computer Science Lab	Undergraduate Researcher	Sep 2019 - Dec 2019
◦ Researched model compression techniques, including structured pruning, regularization-based sparsity, and weight quantization, to reduce the computational footprint of deep convolutional networks. ◦ Applied regularization and pruning strategies to ResNet, achieving a 15% reduction in FLOPs with a 2% improvement in accuracy, demonstrating efficient compression without performance trade-offs.		

PROJECTS

- Nano-Reasoner: Unified Post-Training Framework for Math Reasoning Models [\[github\]](#)
- Designed an end-to-end research-grade, memory-efficient **post-training** pipeline for **reasoning models** with **SFT** and **reinforcement learning** in PyTorch. Implemented cutting-edge reinforcement learning algorithms (**PPO**, **GRPO**, **Dr. GRPO**, **GSPO**, **DAPO**), achieving **20% accuracy wins** in math reasoning benchmarks.
- PRM-Math: Inference-Time Compute Scaling via Dense Process Supervision [\[github\]](#)
- Fine-tuned a **Process Reward Model** in PyTorch to assign dense, step-wise scores to reasoning trajectories.
 - Implemented cutting-edge **test-time compute scaling** strategies (**Best-of-N Search**, **PRM-Weighted Voting**, and **Monte Carlo Tree Search**), resulting in a **10% accuracy improvement** in math reasoning benchmarks.
 - Optimized inference loops with vLLM custom kernels, **reducing latency by 40%**.
- Nano-Video-Gen: Spacetime Diffusion Transformer with Rectified Flow Matching [\[github\]](#)
- Engineered a **Diffusion Transformer video generator** from scratch in PyTorch with **3D spacetime patch embeddings** and **Rectified Flow Matching**. Achieved convergence on Moving MNIST within 48 hours on a single GPU by optimizing the Flash Attention mechanism and implementing mixed-precision training.
- Agentic Deep Research: Recursive Reasoning and Self-Correction Engine [\[github\]](#)
- Designed a **recursive research agent** in LangGraph that executes **dynamic DAGs** for complex, multi-hop query resolution and utilizes **cyclic self-correction loops** to refine search queries. Built a **multi-agent orchestration layer** where sub-agents parallelize retrieval, synthesis, and critique, significantly reducing hallucination rates.
 - Developed a rigorous **LLM-as-a-Judge evaluation** pipeline, demonstrating improvements in **perfect scores (+14%)**, **factuality (+14%)**, **completeness (+3%)**, and **coherence (+2%)** over naive RAG baselines.
- Video-DPO: Temporal Alignment for Video Diffusion via Direct Preference Optimization [\[github\]](#)
- Implemented a **synthetic data** pipeline and adapted **DPO for video diffusion** models to align generation for temporal stability and motion smoothness, **reducing temporal warping artifacts by 25%**.
- Visual-CoT: Pixel-Grounded Reasoning with Multi-Modal Chain-of-Thought [\[github\]](#)
- Fine-tuned a **Vision-Language Model** to generate **interleaved reasoning traces** containing both text and precise bounding box coordinates, forcing the model to spatially attend to evidence. Demonstrated a significant **reduction in object hallucination** and improved **interpretability on complex visual reasoning tasks**.
- Tool-Use DPO: Schema-Constrained Alignment via Identity Preference Optimization [\[github\]](#)
- Fine-tuned an **LLM** for **rigid API contract adherence** using **IPO**, offering greater stability than standard DPO for deterministic outputs. Engineered a **synthetic dataset** to penalize subtle schema violations, **reducing syntax errors by 16%** on complex nested JSON generation tasks.

PUBLICATIONS

- FlapAI Bird: Training an Agent to Play Flappy Bird Using Reinforcement Learning Techniques
Tai Vu, Leon Tran [\[paper\]](#) [\[github\]](#)
- How Not to Give a FLOP: Combining Regularization and Pruning for Efficient Inference
Tai Vu, Emily Wen, Roy Nehoran [\[paper\]](#) [\[github\]](#)
- Privacy Preserving Inference of Personalized Content for Out of Matrix Users
Michael Sun, Tai Vu, Andrew Wang [\[paper\]](#) [\[github\]](#)
- GANime: Generating Anime and Manga Character Drawings from Sketches with Deep Learning
Tai Vu, Robert Yang [\[paper\]](#) [\[github\]](#)
- BERT-VQA: Visual Question Answering on Plots
Tai Vu, Robert Yang [\[paper\]](#) [\[github\]](#)
- Pixel-Perfect Piloting: Superhuman Control of Pixelcopter via Reinforcement Learning
Tai Vu, Brad Nikkel, Jenny Yang [\[paper\]](#) [\[github\]](#)

SKILLS

Languages: Python • C • C++ • SQL • JavaScript • TypeScript • Java • R • PHP
Technologies: PyTorch • JAX • TensorFlow • HuggingFace • NumPy • Pandas • vLLM • Accelerate • FSDP
TRL • LangGraph • OpenRLHF • Unsloth • Scikit-learn • Spark • React Native • HTML • CSS • Express.js
Node.js • React • MongoDB • Next.js • Django
Coursework: Machine Learning • Deep Learning • LLMs • NLP • Reinforcement Learning • Computer Vision
Information Retrieval • Spoken Language Processing • Graph ML • Parallel Computing • Data Structures & Algorithms • Systems • Web Applications • Statistics • Linear Algebra • Convex Optimization • Game Theory